

Karan M

SR GCP Data Engineer

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Professional Summary:

- 5+ years of professional experience in Data Systems Development, focusing on Data Engineering with GCP, AWS.
- Proficiently engaged in all phases of the Software Development Life Cycle (SDLC) and actively participated in daily Scrum meetings, fostering collaboration across cross-functional teams.
- Demonstrated an excellent understanding of AWS, encompassing AWS S3 and EC2, and have leveraged these cloud services effectively.
- Consider me an expert in constructing Enterprise Data Warehouses and Data Warehouse appliances from the ground up, adept in both Kimball and Inman methodologies.
- Hands-on experience in importing and exporting data using stream processing platforms, including Flume and Kafka, ensuring efficient data movement.
- Effectively utilized SSIS for data integration and SSRS for creating and managing reports, providing valuable insights to organizations.
- Monitored and optimized AWS resources with CloudWatch, ensuring high availability and performance of data solutions.
- Orchestrated data workflows using AWS Step Functions, enabling seamless integration between different AWS services.
- Employed AWS RDS for relational database management, ensuring high availability and scalability.
- Extensive experience in building Storm topologies to ingest data from Kafka producers and deliver it to Cassandra databases, ensuring the real-time processing of streaming data.
- Proficient in hosting applications on Google Cloud Platform (GCP), leveraging Compute Engine, App Engine, Cloud SQL, Kubernetes Engine, and Cloud Storage for seamless deployment.
- Developed and deployed various Lambda functions in AWS, both with built-in AWS Lambda Libraries and custom libraries in Scala.
- Setup monitoring and alerting systems using Stackdriver to monitor Dataflow pipelines, ensuring high availability, reliability, and performance of data processing tasks.
- Implemented security best practices for Dataproc clusters, configuring network firewall rules, enabling encryption for data at rest and in transit, and implementing access controls to ensure data privacy and compliance with security standards.
- Optimized Cloud Build configurations to minimize build costs by leveraging caching mechanisms, parallelizing build steps, and optimizing resource allocation based on build requirements, resulting in cost savings while maintaining fast build times.
- Implemented monitoring and analytics solutions to track the performance and usage of data products, analyzing user behavior and feedback to identify areas for improvement and optimization.
- Skilled in implementing automatic Continuous Integration and Continuous Delivery (CI/CD) pipelines using Jenkins. This includes deploying microservices to AWS ECS, running Python jobs in AWS Lambda, and containerized deployments of Java and Python.
- Competent in working with Terraform, making use of its infrastructure as code, execution plans, and change automation features. Also, I have hands-on experience in Kubernetes for cluster management in AWS.
- Engaged with Big Data tools on Google Cloud Platform, including BigQuery, Pub/Sub, Dataproc, and Dataflow, facilitating large-scale data processing and analysis.
- Proficiently used data modeling tools such as Erwin, Power Designer, and ER Studio. Additionally, I have hands-on experience with Hadoop, HDFS, and Hive.
- Strong experience in data migration, cleansing, transformation, integration, import, and export. This involves efficiently moving data between systems and platforms.

- Excellent technical and analytical skills, with a clear understanding of design objectives. This proficiency extends to designing for Online Transaction Processing (OLTP) and dimensional modeling for Online Analytical Processing (OLAP).
- Extensive knowledge of Snowflake Database, its schema, and table structures.
- Worked on handling micro-batching for ingesting numerous files into Snowflake cloud, allowing for seamless data processing when files arrive in the staging area.
- Have actively worked on Scala-based Apache Spark projects, involving RDD transformations, Data Frame operations, and Spark SQL, as well as Spark Streaming.
- Proficient in Snowflake utilities, including SnowSQL and SnowPipe, and have used Python and Java for implementing Big Data model techniques.
- Demonstrated expertise in migrating data warehouses and databases to Hadoop and No SQL platforms, ensuring seamless transitions.
- Designed and developed Oracle PL/SQL scripts and Shell Scripts, specializing in data conversions and data cleansing.
- Proficiency in query optimization and ETL (Extract, Transform, Load) loading on Teradata.
- Extensive experience working with ETL tools, including Teradata Utilities, Informatica, and Oracle.
- Understanding of Apache Spark job execution components, including DAG (Directed Acyclic Graph), lineage graph, DAG Scheduler, Task Scheduler, and Cron Tab Stages. Worked on NoSQL databases like HBase and Mongo DB.
- Created insightful dashboards using Tableau and Power BI to fulfill various business requirements and stakeholder needs.
- Proficient in Python scripting, including statistical functions with NumPy and data visualization using Matplotlib and Pandas.
- Experience in using SQOOP for importing and exporting data from RDBMS to HDFS and Hive. Knowledgeable about job workflow scheduling and tools/services like Oozie, Zookeeper, Airflow, and Apache NiFi.
- Familiar with Amazon Web Service (AWS) concepts like EMR (Elastic MapReduce) and EC2 web services. Successfully loaded files to Hadoop Distributed File System (HDFS) from Oracle, SQL Server, Teradata, and Netezza using SQOOP.
- Evaluated and addressed reporting requirements across various business units, offering data-driven insights.
- Proficient in data preparation, data modeling, and data visualization using Power BI, including the development of various analysis services using Data Analysis Expressions (DAX) queries.
- Demonstrated the ability to address complex Proof of Concepts (POCs) based on business requirements and develop comprehensive test cases for unit testing.

Technical Skills:

Hadoop/Big Data Technologies	Hadoop, Map Reduce, Sqoop, Hive, Oozie, Spark, Zookeeper and Cloudera Manager, Kafka, Flume
ETL Tools	Informatica
Cloud technologies:	Google Cloud Platform (GCP), Amazon Web Services (AWS).
NO SQL Database	HBase, Cassandra, Dynamo DB, Mongo DB.
Monitoring and Reporting	Tableau, Custom shell scripts
Hadoop Distribution	Horton Works, Cloudera
Build Tools	Maven
Programming & Scripting	Python, Scala, JAVA, SQL, Shell Scripting, C, C++
Databases	Oracle, MY SQL, Teradata
Version Control	Git, GitHub, SVN, CVS
Operating Systems	Linux, Unix, Mac OS-X, CentOS, Windows 10, Windows 8, Windows 7

Professional Experience:

Client: CVS Pharmacy, MA.
Role: GCP Data Engineer

April 2022- Present

Responsibilities:

- Migration of an entire Oracle database to BigQuery and employed Power BI for robust reporting. Crafted data pipelines within Google Cloud's Airflow to streamline ETL tasks using a variety of airflow operators.
- Proficient in a range of Google Cloud Platform (GCP) services including Dataproc, Google Cloud Storage (GCS), Cloud Functions, and BigQuery.
- Implemented DAGs (Directed Acyclic Graphs) in Apache Airflow to orchestrate and automate complex workflows, incorporating tasks such as data ingestion, transformation, and loading (ETL).
- Configured Apache Airflow to schedule and monitor data pipelines, ensuring timely execution and reliability of data workflows.
- Demonstrated expertise in Google Cloud components, Google Container Builders, GCP client libraries, and Cloud SDKs.
- Developed and optimized Snowflake data models to support various analytical and reporting requirements, ensuring efficient data storage and retrieval.
- Implemented Snowflake's semi-structured data capabilities to handle and analyze complex JSON data structures, improving data processing efficiency by 30%.
- Designed and implemented ETL pipelines using Snowflake Snowpipes and Stream Ingestion, enabling real-time data loading and processing for critical business dashboards.
- Actively participated in migrating on-premises Hadoop systems to GCP (Google Cloud Platform).
- Conducted in-depth analysis of data from diverse domains to enable seamless integration into a Data Marketplace.
- Developed Pyspark programs, established data frames, and executed data transformations.
- Proficiently employed a variety of GCP services, including GCP Cloud Storage, Dataproc, Data Flow, BigQuery, EMR, S3, Glacier, and EC2 with EMR clusters.
- Solid experience working with Microsoft Azure services like Data Lake, Data Lake Analytics, SQL Database, Synapse, Data Bricks, Data Factory, Logic Apps, and SQL Data Warehouse, along with GCP services like BigQuery, Dataproc, and Pub/Sub.
- Conducted Tableau training sessions for business users to enhance self-service analytics capabilities, resulting in a 50% increase in dashboard adoption and user engagement.
- Integrated Tableau with various data sources including SQL databases, Salesforce, and flat files, ensuring data accuracy and consistency in visualizations and reports.
- Developed and maintained scalable ETL pipelines using AWS services like Glue, Redshift, and Lambda to handle large data sets efficiently.
- Leveraged AWS S3 for secure and scalable storage of structured and unstructured data.
- Implemented data warehousing solutions using AWS Redshift, optimizing query performance and storage costs.
- Designed and developed data pipelines using Google Dataflow to process and analyze streaming data in real-time, ensuring timely insights for business decision-making.
- Deployed and managed Apache Hadoop and Apache Spark clusters on Google Cloud Dataproc, optimizing cluster configurations for performance and cost efficiency.
- Developed data processing pipelines using Apache Beam to handle large-scale data transformations efficiently.
- Optimized Apache Beam pipelines for performance by implementing parallel processing and leveraging data partitioning techniques.
- Implemented CI/CD pipelines using Google Cloud Build to automate build, test, and deployment processes, enabling rapid and reliable delivery of software applications.
- Led requirements gathering sessions with stakeholders to understand business needs and translated them into technical specifications for data product development, ensuring alignment with organizational goals.
- Proficiently analyzed data using Pyspark and Hive based on ETL mappings.
- Implemented a Continuous Delivery pipeline using tools such as Maven, Ant, Jenkins, and GCP. Experienced with Hadoop versions 2.6.4 and 3.1.5.
- Developed multi-cloud strategies, leveraging the strengths of GCP, especially its Platform as a Service (PaaS) offerings.
- Proficiently migrated legacy systems into GCP's cutting-edge technologies.
- Implemented daily data file storage in Google Cloud S3 buckets, effectively harnessing DataProc and BigQuery for maintaining cloud-based solutions.

- Developed Pyspark scripts to merge static and dynamic files while cleansing the data.
- Collaborated with various business units to steer the design and development strategy. Produced functional specifications and technical design documentation. Coordinated with teams such as cloud security, Identity Access Management, Platform, and Network to secure the necessary accreditations and intake processes.
- Leveraged cloud and GPU computing technologies for the automation of machine learning and analytics pipelines, with a primary focus on GCP.
- Actively engaged in a Proof of Concept (POC) to assess different cloud offerings, including the Google Cloud Platform (GCP).
- Conducted a comparative analysis between self-hosted Hadoop and GCP's DataProc, while also exploring Big Table (managed HBase) use cases and evaluating performance improvements.
- Designed multiple Jenkins jobs to facilitate continuous integration processes, and orchestrated a complete CI/CD pipeline using Jenkins.
- Actively participated in the setup of the Apache Airflow service within GCP.
- Utilized the Cloud Shell SDK in GCP to configure services such as DataProc, Storage, and BigQuery.

Environment: AWS, Pyspark, GCPs Data Proc, Big Query, Hadoop, Hive, GCS, Python, Snowflake, Dynamo DB, Oracle Database, Power BI, SDK'S, Data Flow, Glacier, EC2, EMR Cluster, SQL Database, Synapse, Data Bricks.

Client: Digital Technology, INC Chennai

April 2019- July 2021

Role: AWS Data Engineer

Responsibilities:

- Extensive experience working within the AWS cloud platform, with a strong grasp of AWS services like EC2, S3, EMR, Redshift, Lambda, and Glue.
- Utilized Spark RDD, Data Frame API, Data Set API, Data Source API, Spark SQL, and Spark Streaming alongside SQL and MongoDB for comprehensive data processing.
- Enhanced Apache Airflow functionalities by developing custom operators and sensors tailored to specific integration needs with technologies like AWS S3, Google Cloud Storage, and Apache Spark.
- Managed dependencies and execution order of tasks within Apache Airflow DAGs, optimizing workflow efficiency and minimizing job failures.
- Conducted performance tuning and optimization of SQL queries in Snowflake, reducing query execution time by 40% through query profiling and index optimizations.
- Collaborated with cross-functional teams to migrate on-premises data warehouses to Snowflake cloud environment, ensuring seamless data integration and minimal downtime during the transition.
- Leveraged Apache Airflow's web UI and REST API to monitor pipeline status, track task progress, and troubleshoot issues, ensuring seamless operation and performance optimization.
- Successfully developed and deployed data pipelines in cloud environments, particularly AWS.
- Demonstrated a deep understanding of AWS components such as EC2 and S3.
- Developed Spark applications using both Python and R, including implementing Apache Spark data processing projects to handle data from various RDBMS and streaming sources.
- Employed Apache Spark's data frames, Spark-SQL, and Spark MLlib extensively, designing and developing POCs using Scala, Spark SQL, and MLlib libraries.
- Integrated Apache Beam with various data sources including Apache Kafka and Google Cloud Pub/Sub for real-time data ingestion.
- Designed and implemented error handling and retry mechanisms in Apache Beam pipelines to ensure fault tolerance and data consistency.
- Demonstrated expertise in data integration involving structured data, employing traditional ETL tools and methodologies to collect data from diverse sources into a single data warehouse.
- Involved in the design and deployment of multi-tier applications using AWS services such as EC2, Route53, S3, RDS, Dynamo DB, SNS, SQS, and IAM, focusing on high availability, fault tolerance, and auto-scaling within AWS Cloud Formation.
- Created interactive and visually appealing Tableau dashboards to visualize key business metrics and KPIs, facilitating data-driven decision-making across departments.

- Developed complex Tableau calculations including LOD expressions and table calculations to provide deeper insights into customer behavior and sales trends.
- Implemented Tableau Server for centralized dashboard deployment and management, ensuring secure access and efficient sharing of analytical insights across the organization.
- Automated data ingestion processes with AWS Lambda, reducing manual intervention and improving data flow efficiency.
- Collaborated with cross-functional teams to troubleshoot and resolve issues related to Apache Beam pipelines, ensuring smooth data flow and processing.
- Utilized AWS Glue for data cataloging and transforming data between various formats, ensuring data consistency and quality.
- Designed and managed AWS EMR clusters for big data processing, leveraging Apache Spark for complex data transformations.
- Developed prototypes and minimum viable products (MVPs) for data products using agile methodologies, iteratively refining features based on user feedback and performance metrics.
- Created custom build steps in Cloud Build to perform specialized tasks such as code linting, unit testing, code coverage analysis, and vulnerability scanning, ensuring code quality and security compliance.
- Integrated Dataflow pipelines seamlessly with other Google Cloud Platform services like BigQuery, Pub/Sub, and Cloud Storage to create end-to-end data processing workflows.
- Proficiency in Python and Scala, with a knack for creating user-defined functions (UDF) for Hive and Pig using Python.
- Efficiently extracted data from SQL server, Teradata, Amazon S3 buckets, and internal SFTP, loading them into AWS S3 buckets in a data warehouse context.
- Developed Spark jobs for data processing and orchestrated instances and clusters to load data into AWS S3 buckets, thereby creating a DataMart.
- Utilized AWS EMR for processing and transforming data to assist the Data Science team based on business requirements.
- Designed and developed ETL processes in AWS Glue to migrate campaign data from external sources, such as S3, ORC/Parquet/Text files, into AWS Redshift.
- Worked on both batch processing and real-time data processing using Spark Streaming with a Lambda architecture.
- Developed Spark applications for data cleaning and validation of ingested data within the AWS cloud.
- Crafted MapReduce jobs using Java to process and validate data, ranging from simple to complex tasks.
- Developed Python code for various tasks, dependencies, and time sensors in the context of workflow management and automation using the Airflow tool.
- Developed scripts to load data to Hive from HDFS and participated in data ingestion into Data Warehouses using various data loading techniques.
- Utilized Spark Streaming APIs for on-the-fly data transformations and actions. Developed preprocessing jobs using Spark DataFrames to flatten JSON documents into flat files.
- Employed Kubernetes for the runtime environment of the CI/CD system to facilitate building, testing, and deployment.
- Designed and implemented data access controls and security measures within Google FHIR Store to protect sensitive patient information, ensuring adherence to HIPAA regulations and other healthcare data privacy standards.
- Developed custom data queries and analytics pipelines using Google FHIR Store APIs and BigQuery integration, enabling healthcare organizations to derive insights and support clinical decision-making.
- Collaborated with healthcare domain experts and software engineers to design and deploy scalable architectures leveraging Google FHIR Store, ensuring high availability and reliability of healthcare data services.
- Collaborated with the DevOps team to implement NIFI Pipelines on EC2 nodes, integrated with Spark, Kafka, Postgres running on other instances, using SSL handshakes in QA and Production Environments.
- Developed Informatica mappings, sessions, and workflows and facilitated version control through Informatica's Versioning Control for code changes.

Environment: AWS, EMR, Scala, MapReduce, GCP, HDFS, Hive, Pig, Apache Kafka, Sqoop, Python, Pyspark, Shell scripting, Linux, MySQL Oracle Enterprise DB, Big query, SOLR, Jenkins, Eclipse, Dataflow, Tableau, MySQL.

Client: Scadea Solutions Inc

July 2018- March 2019

Role: Big Data Engineer

Responsibilities:

- Integrated a comprehensive metadata management system, empowering data stewards and business analysts to collaborate effectively. This system encompassed data quality assessment, monitoring, and a data cleansing package builder, facilitating governance to ensure data trustworthiness.
- Upgraded the Hadoop Cluster, transitioning from CDH 3 to CDH 4. Implemented a High Availability Cluster while seamlessly integrating HIVE with existing applications.
- Proficiently analyzed the Hadoop cluster and various big data analytic tools, including Pig, HBase database, and Sqoop.
- Orchestrated the Extract, Transform, Load (ETL) process, moving data from source systems to Azure Data Storage services. This entailed using a blend of Azure Data Factory, T-SQL, Spark SQL, and Azure Data Lake Analytics.
- Managed data transformations between different servers, utilizing tools such as Data Transformation Services (DTS) and SQL Server Integration Services (SSIS) in versions 2005, 2008, and 2013.
- Demonstrated excellence in SQL Server 2005/2008 RDBMS database development, encompassing T-SQL programming. Created and optimized stored procedures, functions, indexes, tables, and views, along with crafting intricate T-SQL code and SQL joins to support applications.
- Devised and fine-tuned ETL processes, utilizing stored procedures for data transformations, alongside T-SQL statements tailored to specific transformation requirements.
- Improved data processing efficiency by optimizing Dataflow jobs through techniques such as parallelization, windowing, and watermarking, resulting in significant reduction in processing time and cost.
- Utilized Dataproc clusters to process large-scale datasets efficiently, leveraging parallel processing capabilities of Hadoop and Spark to accelerate data processing tasks.
- Integrated Cloud Build with Docker to automate container image builds and push them to Google Container Registry, streamlining the deployment of containerized applications on Google Kubernetes Engine (GKE) or other container orchestration platforms.
- Designed scalable and efficient data models and architectures for data products, considering factors such as data volume, velocity, variety, and veracity, to support complex analytics and reporting requirements.
- Designed, developed, and maintained Tableau functional reports according to user specifications, ensuring meaningful data visualization.
- Utilized Google FHIR Store to store and manage healthcare data in compliance with the HL7 FHIR (Fast Healthcare Interoperability Resources) standard, ensuring interoperability and data consistency across systems.
- Implemented data ingestion pipelines to populate Google FHIR Store from various healthcare data sources such as electronic health records (EHR) systems and medical devices, using APIs and data transformation tools.
- Possessed sound knowledge of No SQL databases, including Dynamo DB and Mongo DB.
- Implemented robust data backup and recovery strategies using AWS S3 and Glacier, ensuring data integrity and availability.
- Integrated AWS Kinesis for real-time data streaming and analytics, enabling prompt data-driven decisions.
- Employed AWS IAM for fine-grained access control, ensuring secure data access and compliance with regulations.
- Developed data pipelines using AWS Data Pipeline, automating data workflows and scheduling.
- Configured HBase tables to accommodate various data formats, specifically PII data from diverse portfolios.
- Skillfully integrated and configured Sqoop, crafting scripts to extract data from MySQL into the Hadoop Distributed File System (HDFS).
- Utilized Hive scripts for data transformations, preparing the data for delivery to target systems, enabling data analysts to conduct comprehensive reporting.
- Designed and developed applications using the Spring framework, adhering to Model-View-Controller (MVC) design patterns.
- Demonstrated a commitment to best practices in unit testing, continuous integration and continuous delivery (CI/CD), performance testing, capacity planning, documentation, monitoring, alerting, and incident response.
- Actively participated in client meetings, engaging with subject matter experts to gather essential functional business requirements.

- Seamlessly integrated data quality (DQ) workflows with Data Integration (DI) ETL jobs, incorporating data cleansing and matching logic.
- Pioneered the creation of several Data Warehouses and Data Marts from the ground up. This encompassed the development of conceptual, logical, and physical models, establishment of processes, and definition of enterprise standards and workflows.
- Demonstrated expertise in handling multiple data formats, including JSON, AVRO, and Parquet. Leveraged Storm, Spark, and Pig scripts to load data into NoSQL databases such as MongoDB and Cassandra.
- Generated new Power BI reports and harnessed the capabilities of Microsoft Azure Cloud for publication and pinning of reports.

Environment: Hadoop, AWS , Power BI, Talend ETL Tool, Mongo DB, Hadoop Cluster, Data Warehouse, NO SQL, HDFS, ETL, MY SQL, Spark, Pig.

Education Details:

Bachelors

: - Buisness Administration,
Dr. C.V. Raman University, JAN 2016 - DEC 2018

Masters

: - Science in Information Technology
University of North America, JAN 3 2022 - Jun 21 2023